

Lesson 11 Counting

1. In a college, the student ID number consists of 7 digits.
 - a) How many possible ID numbers are there?
 - b) How many ID numbers have distinct (different) digits?
 - c) How many ID numbers are odd?
 - d) How many ID numbers have only even digits?
 - e) How many ID numbers alternate between even and odd digits?
 - f) How many ID numbers start with 0 and end with 0?
 - g) How many ID numbers have no 0's?
 - h) How many ID numbers have exactly one 0?
 - i) How many ID numbers have at most two 0's?
 - j) How many ID numbers have at least one 0?
 - k) How many ID numbers are multiples of 5?
 - l) How many ID numbers read the same backward and forward?
 - m) How many ID numbers have exactly two prime digits (2, 3, 5, 7)?
2. Two men and three women pose for a group photo, 3 of them will sit in a row of chairs and the rest will stand in a row behind the chairs.
 - a) How many possible arrangements are there?
 - b) How many arrangements are possible if all the women sit?
 - c) How many arrangements are possible if all the men sit?
3. How many ways are there to arrange four couples in 8 chairs so that every couple sits together?
4. How many ways are there to choose a president, a secretary, a treasurer, and 3 committee members in a club of 20 members?
5. There are 15 people in a party. Suppose everyone shakes hand exactly once with everyone else, how many handshakes are there?
6. How many ways are there to divide 12 elements into 4 sets of 3 elements?
7. In a standard deck of 52 playing cards, there are 4 suites: Club, Diamond, Spade, and Heart. Clubs and spades are black, and diamonds and hearts are red. In each suite, there are 13 values: $A, 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K$.
 - a) How many 13-card bridge hands are there?
 - b) How many 13-card bridge hands contain only red cards?
 - c) How many 13-card bridge hands contain at least one black card?
 - d) How many 5-card poker hands contain exactly 3 diamonds?
8. A binary string consists of 0 and 1; a ternary string consists of 0, 1, and 2.
 - a) How many binary strings of length n are there?
 - b) How many binary strings of length n have 0's?
 - c) How many binary strings of length n have exactly k 0's?
 - d) How many ternary strings of length n are there?
 - e) How many ternary strings of length n have 0's?
 - f) How many ternary strings of length n have exactly k 0's?